

Daksh — Class X Study Test

Subject: Mathematics | **Test:** Test 2 (Day 14, 23 May 2026) — 2-Week Cumulative

Time Allowed: 3 Hours • **Maximum Marks:** 80

Syllabus covered: Chs 1–13: Real Numbers • Polynomials • Linear Equations in Two Variables • Quadratic Equations • AP • Triangles • Coordinate Geometry • Trigonometry • Applications of Trigonometry • Circles • Areas Related to Circles • Surface Areas and Volumes • Statistics

General Instructions:

- (i) This question paper contains five sections — A, B, C, D and E.
- (ii) Section A has Multiple Choice / Very Short Answer Questions of 1 mark each.
- (iii) Section B has Short Answer Questions of 2 marks each.
- (iv) Section C has Short Answer Questions of 3 marks each.
- (v) Section D has Long Answer Questions of 5 marks each.
- (vi) Section E has Case-based / Source-based questions of 4 marks each.
- (vii) All questions are compulsory. Internal choice has been provided in some questions wherever indicated.
- (viii) Use of calculators is not permitted.

SECTION A — Multiple Choice / Very Short Answer Questions (20 × 1 = 20 marks)

Choose the correct option / answer.

- Q1.** The HCF of two co-prime numbers is: (a) 0 (b) 1 (c) the smaller (d) the larger
- Q2.** If α and β are zeroes of $x^2 + 7x + 10$, then $\alpha^2 + \beta^2 = ?$ (a) 9 (b) 29 (c) 39 (d) 49
- Q3.** The pair $x - y = 2$, $3x - 3y = 6$ has: (a) unique solution (b) no solution (c) infinitely many solutions (d) two solutions
- Q4.** The roots of $x^2 - 4x + 4 = 0$ are: (a) real and distinct (b) real and equal (c) imaginary (d) cannot say
- Q5.** The 11th term from the end of the AP: 7, 10, 13, ..., 184 is: (a) 154 (b) 157 (c) 160 (d) 161
- Q6.** If $\triangle ABC \sim \triangle DEF$ and $AB/DE = 2/3$, then $ar(ABC)/ar(DEF) = ?$ (a) $2/3$ (b) $4/9$ (c) $9/4$ (d) $3/2$
- Q7.** The distance between the points $(-5, 7)$ and $(-1, 3)$ is: (a) $4\sqrt{2}$ (b) 4 (c) 8 (d) $2\sqrt{2}$
- Q8.** The midpoint of $A(-4, 6)$ and $B(2, 8)$ is: (a) $(-1, 7)$ (b) $(-2, 7)$ (c) $(3, 14)$ (d) $(-3, 14)$
- Q9.** $\sin 30^\circ + \cos 60^\circ = ?$ (a) 1 (b) $1/2$ (c) $\sqrt{3}/2$ (d) 0
- Q10.** If $\sec \theta = 5/4$, then $\tan \theta = ?$ (a) $3/4$ (b) $4/3$ (c) $3/5$ (d) $5/3$
- Q11.** The angle of elevation of the top of a tower from a point 30 m away on level ground is 30° . The height of the tower is: (a) $10\sqrt{3}$ m (b) $30\sqrt{3}$ m (c) $30/\sqrt{3}$ m (d) 15 m
- Q12.** The number of tangents that can be drawn from a point outside a circle is: (a) 0 (b) 1 (c) 2 (d) infinite
- Q13.** If two tangents drawn to a circle from an external point are inclined at 60° to each other, the angle between the radii at the points of contact is: (a) 60° (b) 90° (c) 120° (d) 150°
- Q14.** The area of a quadrant of a circle of radius 14 cm is: (a) 154 cm^2 (b) 308 cm^2 (c) 77 cm^2 (d) 49 cm^2
- Q15.** A solid sphere of radius r is melted into a cylinder of the same radius. The height of the cylinder is: (a) $(4/3)r$ (b) $(2/3)r$ (c) r (d) $(1/3)r$
- Q16.** The mean of the first 10 prime numbers is: (a) 12.9 (b) 12.5 (c) 12.0 (d) 13.0
- Q17.** Express $0.4\overline{4}$ (0.4 recurring) as a rational number in lowest form.
- Q18.** Find the value of k for which $x^2 + kx + 9 = 0$ has equal roots.
- Q19.** Write the empirical relation between mean, median and mode.
- Q20.** A circle has radius 7 cm. Find its circumference.

SECTION B — Short Answer-I (5 × 2 = 10 marks)

Show all working.

- Q21.** Find the values of x and y if $217x + 131y = 913$ and $131x + 217y = 827$.
- Q22.** Determine an AP whose third term is 16 and the seventh term exceeds the fifth term by 12.

Q23. Find the coordinates of the point which divides the line segment joining (4, -3) and (8, 5) in the ratio 3 : 1 internally.

Q24. Prove that: $(\sec \theta - \tan \theta)(\sec \theta + \tan \theta) = 1$.

Q25. A chord 12 cm long is drawn in a circle of radius 10 cm. Find the perpendicular distance from the centre to the chord.

SECTION C — Short Answer-II ($6 \times 3 = 18$ marks)

Answer with full working.

Q26. Prove that $3 + 2\sqrt{5}$ is irrational, given that $\sqrt{5}$ is irrational.

Q27. If one zero of the polynomial $2x^2 - 3x + p$ is the reciprocal of the other, find p.

Q28. The taxi fare in a city is as follows: for the first kilometre the fare is ₹8, and for subsequent distance it is ₹5 per km. Taking the distance covered as x km and total fare as ₹y, write a linear equation. How much does a person pay for travelling 16 km?

Q29. If the sum of the first n terms of an AP is given by $S_n = 3n^2 + 2n$, find the nth term and hence find the 25th term.

Q30. Prove the identity: $(1 + \cot^2 \theta) \sin^2 \theta = 1$.

Q31. A piece of wire 20 cm long is bent into the form of an arc of a circle subtending an angle of 60° at its centre. Find the radius of the circle.

SECTION D — Long Answer ($4 \times 5 = 20$ marks)

Answer in detail with neat figures wherever needed.

Q32. The angle of elevation of the top of a building from the foot of a tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60° . If the tower is 50 m high, find the height of the building.

Q33. Prove: 'In a right triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.' (Pythagoras Theorem)

Q34. A cone of height 24 cm and base radius 6 cm is made of solid wood. From it, a hemisphere of the same radius as the base of the cone is hollowed out. Find the volume of the remaining solid.

Q35. The following table gives the daily wages of 50 workers in a factory:

Daily wage (₹)	No. of workers
100–120	12
120–140	14
140–160	16
160–180	18
180–200	20

Find the mean, median and mode of the data.

SECTION E — Case-based / Source-based ($3 \times 4 = 12$ marks)

Read each case study and answer the sub-parts.

Q36. Case Study (Coordinate Geometry). A school is constructing a triangular flowerbed in the playground. Three corners of the triangular bed are at points A(2, 1), B(5, 4) and C(2, 4) (coordinates in metres).

- (i) Find the length of side AC. (1)
- (ii) Find the length of side BC. (1)
- (iii) Find the area of the triangular bed. (2)

Q37. Case Study (Trigonometry / Heights & Distances). An aeroplane at an altitude of 1500 m observes the angles of depression of two ships sailing in the same direction at 60° and 30° respectively.

- (i) Draw a labelled figure of the situation. (1)
- (ii) Find the distance between the ships. (3)

Q38. Case Study (Mensuration). A toy is in the form of a cone of base-radius 3.5 cm mounted on a hemisphere of the same radius. The total height of the toy is 15.5 cm.

- (i) Find the height of the cone. (1)
- (ii) Find the slant height of the cone. (1)
- (iii) Find the total surface area of the toy. (Use $\pi = \frac{22}{7}$) (2)